Summary in Graph

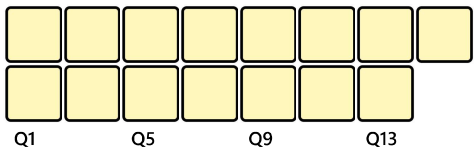
Exam Summary (GO Classes DA Test Series 2026 | Machine Learning | Topic Wise Test 7)

Qs. Attempted:	1 (1 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	1 (1 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (10 + 5)
Total Marks:	20 (10 + 10)
Exam Duration:	45 Minutes
Time Taken:	45 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (32) out of 322 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

Which of the following is/are always true about the ROC curve and the area under the ROC curve (AUC) for a binary classifier?

A. ROC is obtained by varying the the threshold of the classifier.
B. For a classifier with $\mathrm{AUC} = 0.5$, the performance is worse than the random guess (no power) classifier.
C. The ROC curve increases monotonically with the false positive rate.
D. The ROC curve explains the tradeoff between precision and specificity.

Your Answer: A;D | Correct Answer: A;C | Incorrect | Time taken: 14min 47sec | Discuss

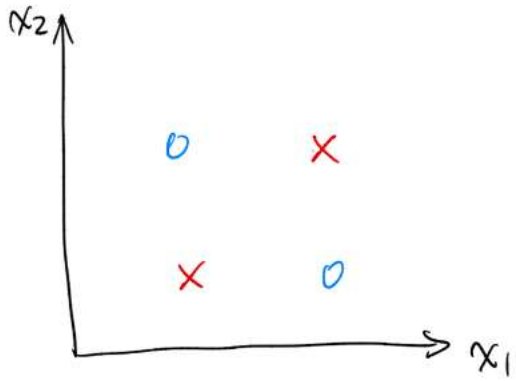
Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider building a spam classifier (binary: spam vs non-spam) for your mailbox. If we assume spam to be the positive class, which of the following would be more important to optimize? Note that we do not want to classify genuine emails as spam emails.

A. Precision
B. Recall
C. Both Precision and Recall
D. Accuacy.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 34min 50sec | Discuss

Q #3 | Numerical Type | Award: 1 | Penalty: 0 | Machine Learning



Each point is an example in $\mathbb{R}^2$, i.e. $x = (x_1, x_2)$. Circles are positive examples and crosses are negative examples.
What's the maximum training accuracy we can achieve on this dataset if we use a linear classifier? Provide the answer in percentage.

Your Answer: | Correct Answer: 75 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

consider a binary classification data set with 9000 positively labelled examples and 1000 negatively labelled examples. What is the precision and the recall of a classifier that always classifies any example as positive?

A. The precision is 0.1 , and the recall is 0.9 .
B. The precision is 0.9 , and the recall is 0.1 .
C. The precision is 1.0 , and the recall is 0.9 .
D. The precision is 0.9 , and the recall is 1.0 .

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

For a probabilistic classifier used in a binary classification problem, consider the prediction rule: predict class 1 if $\mathrm{P}(\mathrm{y} = 1 \mid x) > \mathrm{t}$, and predict class 0 otherwise. This rule assumes a threshold t is set.
What happens when the threshold t is increased?

A. Precision increases, and recall decreases.
B. Precision decreases, and recall increases.
C. Both precision and recall increase.
D. Both precision and recall decrease.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Increasing the regularization of a model would typically:

A. Increase its bias and increase its variance
B. Increase its bias and decrease its variance
C. Decrease its bias and increase its variance
D. Decrease its bias and decrease its variance

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Suppose your model is demonstrating high variance across different training sets. Which of the following is NOT a valid way to try and reduce the variance?

A. Increase the amount of training data in each training set.
B. Improve the optimisation algorithm being used for error minimisation.
C. Decrease the model complexity.
D. Reduce the noise in the training data.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider a random variable X with unknown mean μ and unknown variance σ^2. Given n iid realizations of training samples $x_1, x_2, x_3, \ldots, x_n$ from the random variable, we wish to estimate the mean of X. Our estimate of μ is the random variable $\hat{X}$, which has mean $\hat{\mu}$. We estimate μ using the average of the n samples:

$$\frac{x_1 + x_2 + \ldots + x_n}{n}$$

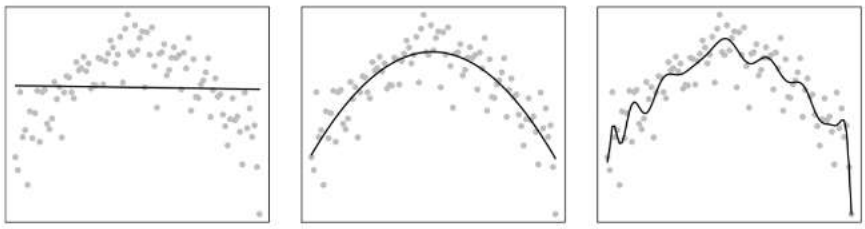
What is the bias of the estimator above?

A. The bias is μ
B. The bias is $\frac{\mu}{n}$
C. The bias is 0
D. The bias is $-\mu$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

Consider the three line regression fits to the same gray points plotted below.



$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 x$ (1)

$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 x + \hat{\beta}_2 x^2$ (2)

$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 x + \cdots + \hat{\beta}_{20} x^{20}$ (3)

Which of the following statements is/are correct?

A. The estimate in (2) has a higher variance than the estimate in (1).
B. The estimate in (2) has a higher bias than the estimate in (3).
C. The estimate in (3) has the smallest training error among the three.
D. The estimate in (1) has the smallest test error among the three.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider the error decomposition for a least squares regression model

$$\mathbf{E}_{x,y,D}\left[(h(x;D)-y)^2\right] = \underbrace{\mathbf{E}_{x,D}\left[(h(x;D)-\bar{h}(x))^2\right]}_{\text{Variance}} + \underbrace{\mathbf{E}_{x}\left[(\bar{h}(x)-\bar{y}(x))^2\right]}_{\text{Bias}^2} + \underbrace{\mathbf{E}_{x,y}\left[(\bar{y}(x)-y)^2\right]}_{\text{Noise}}$$

where $h(x;D)$ is a model learned over a training sample D, $\bar{h}(x) = \mathbf{E}_{D\sim P^n}[(h(x;D)]$ is the average model, and $\bar{y}(x) = \mathbf{E}_{y|x}[y]$ is the optimal Bayes model. Which of the following best describes the term labeled bias ?

A. On average, how much your learned model differs from average model across different samples D
B. How far is the average model from optimal model
C. The variance between predictions for a fixed sample D
D. How accurate the model is in predicting y

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider a binary classification problem with two outcomes: positive and negative. The F_1 score of a classifier in such a problem is the harmonic mean between its precision and recall:

$$F_1 = 2\frac{p \cdot r}{p + r}$$

Select the true statement relating a classifier's F_1 score on the test set to the bias and variance of its estimated parameters. You may assume the classifier is operating in the classical (noninterpolating) regime of bias and variance.

A. Low bias implies a high F_1 score.
B. Low variance implies a high F_1 score.
C. A low F_1 score implies a high variance.
D. A high F_1 score implies low bias.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Given a design matrix $X \in \mathbb{R}^{n \times d}$, labels $y \in \mathbb{R}^n$, and $\lambda > 0$, we find the weight vector w^* that minimizes $\|Xw - y\|^2 + \lambda \|w\|^2$. Suppose $w^* \neq 0$.
Which of the following statements is/are correct?

A. The variance of the method decreases if λ increases enough.
B. There may be multiple solutions for w^*.
C. The bias of the method increases if λ increases enough.
D. The variance of a model typically decreases as the number of features increases.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Which of the following statements regarding ROC curves are true?

A. The ROC curve is monotonically increasing
B. for a logistic regression classifier, the ROC curve's horizontal axis is the posterior probability used as a threshold for the decision rule
C. The ROC curve is concave
D. if the ROC curve passes through $(0, 1)$, the classifier is always correct (on the test data used to make the ROC curve)

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

We will analyze the model bias and variance of a simple linear regression model with no intercept term, given by $f_\theta(x) = \theta x$. Note that both x and θ are 1 -dimensional.
Suppose we estimate the values on the right using our model $f_{\hat{\theta}}(x)$ on the test data point $(x, Y) = (2, 3)$.
Given the following quantities:

Quantity	Value
$\mathrm{E}[\mathrm{f}_{\hat{\theta}}(\mathrm{x})]$	1
$\mathrm{E}[(\mathrm{f}_{\hat{\theta}}(\mathrm{x}))^2]$	6
σ^2	0

What is the model risk, assuming there is no observational variance $(\epsilon = 0)$?

A. 5
B. 6
C. 9
D. 4

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Using a simple linear regression model without an intercept (i.e., $f_\theta(x) = \theta x$), we estimate the model variance at a new data point $(x, Y) = (3, 12)$ as $\mathrm{Var}\big(f_{\tilde{\theta}}(x)\big) = 3$ using some arbitrary model parameter $\tilde{\theta}$.
Suppose we create a new model where $\theta_{\text{new}} = 2\tilde{\theta}$.
What is the new model variance, $\mathrm{Var}(f_{\theta_{\text{new}}}(x))$, using θ_{new} ?

A. 6
B. 9
C. 12
D. 15

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

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